

Coherence-Modulated Gravity in the Maat Field Theory of Everything

Christof Krieg

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Abstract

We present a compact and reproducible pipeline exploring a single-field slice of the Maat Field Theory of Everything (MFToE) with a dynamic vacuum controller field χ . The model is integrated numerically in $N = \ln a$ and compared with baryon acoustic oscillation (BAO) measurements from DESI Data Release 2.

We find that the MFToE background evolution produces sub-percent deviations in the Hubble expansion rate and luminosity distance relative to Λ CDM while remaining consistent with current BAO constraints.

Using the observed residuals, we perform a first phenomenological fit of the coherence coupling and obtain

$$\xi A \approx 1.6 \times 10^{-3}, \quad (1)$$

indicating that the MFToE coherence sector naturally operates in the sub-percent regime relevant for observational cosmology.

These results suggest that the MFToE framework can reproduce late-time cosmological observations while allowing small dynamical departures from Λ CDM driven by a vacuum self-regulation mechanism.

In addition, we position the MFToE framework within the landscape of scalar–tensor gravity by comparing it to Brans–Dicke and Horndeski-type models. A linear stability analysis of the vacuum controller field χ shows that the late-time attractor remains stable for positive effective mass $m_{\chi, \text{eff}}^2 > 0$, with Hubble friction providing additional damping. This indicates that the coherence-controlled vacuum sector can produce small cosmological deviations without destabilizing the background expansion.

1 Coherence-Modulated Gravity

To sharpen the gravitational sector of the Maat Field Theory of Everything (MFToE), we introduce a coherence-modulated scalar–tensor extension of General Relativity. The central idea is to distinguish between the elementary Maat fields

$$\phi_H, \phi_B, \phi_S, \phi_V, \phi_R$$

and a derived macroscopic coherence function that summarizes their joint state.

1.1 Coherence Function

We define the local coherence scalar as

$$\mathcal{C}(x) = \frac{\phi_H(x)\phi_B(x)\phi_S(x)\phi_V(x)\phi_R(x)}{\Delta E(x) + \epsilon}, \quad (2)$$

where $\Delta E(x)$ denotes the local deviation from energetic equilibrium and $\epsilon > 0$ is a regulator preventing singular behavior.

1.2 Extended Action

We propose the action

$$S = \int d^4x \sqrt{-g} \left[\frac{M_{\text{Pl}}^2}{2} F(\mathcal{C}) R - \frac{1}{2} \sum_{i \in \{H, B, S, V, R\}} g^{\mu\nu} \partial_\mu \phi_i \partial_\nu \phi_i - U(\phi_i, \mathcal{C}) + \mathcal{L}_m \right], \quad (3)$$

with

$$F(\mathcal{C}) = 1 + \xi \mathcal{C}, \quad (4)$$

and potential

$$U(\phi_i, \mathcal{C}) = \lambda \left(\frac{\prod_i \phi_i}{\Delta E + \epsilon} - M_0 \right)^2 + \frac{1}{2} \sum_{i,j} G_{ij} \phi_i \phi_j. \quad (5)$$

2 Field Equations

Variation with respect to the metric yields the modified Einstein equations

$$F(\mathcal{C}) G_{\mu\nu} = \frac{1}{M_{\text{Pl}}^2} \left(T_{\mu\nu}^{(m)} + T_{\mu\nu}^{(\phi)} \right) + \nabla_\mu \nabla_\nu F(\mathcal{C}) - g_{\mu\nu} \square F(\mathcal{C}), \quad (6)$$

where

$$T_{\mu\nu}^{(\phi)} = \sum_i \left(\partial_\mu \phi_i \partial_\nu \phi_i - \frac{1}{2} g_{\mu\nu} \partial_\alpha \phi_i \partial^\alpha \phi_i \right) - g_{\mu\nu} U(\phi_i, \mathcal{C}). \quad (7)$$

Variation with respect to ϕ_k gives

$$\square \phi_k - \frac{\partial U}{\partial \phi_k} + \frac{M_{\text{Pl}}^2}{2} F'(\mathcal{C}) R \frac{\partial \mathcal{C}}{\partial \phi_k} = 0. \quad (8)$$

The derivative of the coherence scalar is

$$\frac{\partial \mathcal{C}}{\partial \phi_k} = \frac{1}{\Delta E + \epsilon} \prod_{i \neq k} \phi_i. \quad (9)$$

3 FLRW Reduction

For a spatially flat FLRW background

$$ds^2 = dt^2 - a(t)^2 d\vec{x}^2,$$

and homogeneous fields $\phi_i = \phi_i(t)$, the coherence scalar becomes

$$\mathcal{C}(t) = \frac{\phi_H(t) \phi_B(t) \phi_S(t) \phi_V(t) \phi_R(t)}{\Delta E(t) + \epsilon}. \quad (10)$$

The modified Friedmann equation reads

$$3M_{\text{Pl}}^2 F(\mathcal{C}) H^2 = \rho_m + \rho_r + \rho_\phi - 3M_{\text{Pl}}^2 H \dot{F}(\mathcal{C}), \quad (11)$$

with

$$\rho_\phi = \sum_i \frac{1}{2} \dot{\phi}_i^2 + U(\phi_i, \mathcal{C}). \quad (12)$$

The acceleration equation is

$$-2M_{\text{Pl}}^2 F(\mathcal{C}) \dot{H} = (\rho_m + p_m) + (\rho_r + p_r) + \sum_i \dot{\phi}_i^2 + M_{\text{Pl}}^2 \ddot{F}(\mathcal{C}) - H M_{\text{Pl}}^2 \dot{F}(\mathcal{C}). \quad (13)$$

For each homogeneous field,

$$\ddot{\phi}_k + 3H \dot{\phi}_k + \frac{\partial U}{\partial \phi_k} - \frac{M_{\text{Pl}}^2}{2} F'(\mathcal{C}) R \frac{\partial \mathcal{C}}{\partial \phi_k} = 0. \quad (14)$$

4 Limit to General Relativity and Λ CDM

The theory must reduce to standard cosmology in an appropriate limit. This occurs when the coherence modulation becomes negligible,

$$\xi\mathcal{C} \ll 1, \quad (15)$$

such that

$$F(\mathcal{C}) \rightarrow 1. \quad (16)$$

In this regime, the gravitational sector reduces to the Einstein–Hilbert form and Eq. (6) becomes

$$G_{\mu\nu} = \frac{1}{M_{\text{Pl}}^2} \left(T_{\mu\nu}^{(m)} + T_{\mu\nu}^{(\phi)} \right). \quad (17)$$

If, in addition, all Maat fields settle near constant equilibrium values,

$$\dot{\phi}_i \approx 0, \quad \phi_i \approx \phi_i^{(0)}, \quad (18)$$

then the potential approaches a nearly constant vacuum contribution,

$$U(\phi_i, \mathcal{C}) \approx U_0, \quad (19)$$

which behaves like an effective cosmological constant. In that limit, the model reduces to Λ CDM with

$$\rho_\phi \approx U_0, \quad p_\phi \approx -U_0, \quad w_\phi \approx -1. \quad (20)$$

Thus, General Relativity and Λ CDM are recovered as the low-dynamics, low-coherence-coupling limit of the MFToE framework.

5 Minimal Single-Field Reduction

For phenomenological studies, it is useful to define a reduced model in which only one effective scalar degree of freedom remains dynamical. We choose the Harmony field ϕ_H as the active component and set

$$\phi_B, \phi_S, \phi_V, \phi_R \rightarrow \text{const.} \quad (21)$$

Then the coherence scalar simplifies to

$$\mathcal{C}_{\text{eff}}(t) = \frac{A \phi_H(t)}{\Delta E(t) + \epsilon}, \quad A \equiv \phi_B^{(0)} \phi_S^{(0)} \phi_V^{(0)} \phi_R^{(0)}. \quad (22)$$

The action becomes

$$S_{\text{eff}} = \int d^4x \sqrt{-g} \left[\frac{M_{\text{Pl}}^2}{2} (1 + \xi \mathcal{C}_{\text{eff}}) R - \frac{1}{2} \partial_\mu \phi_H \partial^\mu \phi_H - U_{\text{eff}}(\phi_H) + \mathcal{L}_m \right], \quad (23)$$

with an effective potential

$$U_{\text{eff}}(\phi_H) = \lambda \left(\frac{A \phi_H}{\Delta E + \epsilon} - M_0 \right)^2 + \frac{1}{2} m_H^2 \phi_H^2. \quad (24)$$

This single-field truncation connects the coherence-modulated gravity model directly to quintessence-like dark-energy dynamics while preserving the conceptual origin of the coherence scalar.

6 Low-Redshift Observational Sector

The leading observational consequences of coherence-modulated gravity are expected at late times, where the scalar sector evolves slowly and may mimic dynamical dark energy. The most relevant observables are:

- the Hubble expansion rate $H(z)$,
- the luminosity distance $d_L(z)$,
- the effective dark-energy equation of state $w_\phi(z)$,
- the effective gravitational coupling $G_{\text{eff}}(z)$.

From Eq. (4), the effective gravitational strength is

$$G_{\text{eff}}(z) = \frac{G}{F(\mathcal{C}(z))} = \frac{G}{1 + \xi\mathcal{C}(z)}. \quad (25)$$

At low redshift, the theory predicts small deviations from Λ CDM of the schematic form

$$\frac{\Delta H}{H_{\Lambda\text{CDM}}}(z) \sim \mathcal{O}(\xi\mathcal{C}(z)), \quad \frac{\Delta d_L}{d_{L,\Lambda\text{CDM}}}(z) \sim \mathcal{O}(\xi\mathcal{C}(z)). \quad (26)$$

This makes the low-redshift sector the natural first test bed of the model, particularly using:

- supernova distance data,
- BAO measurements,
- growth-of-structure probes,
- standard sirens and gravitational-wave damping.

A practical next step is therefore to fit the reduced single-field model against low-redshift datasets and compare its performance to Λ CDM and standard quintessence parameterizations.

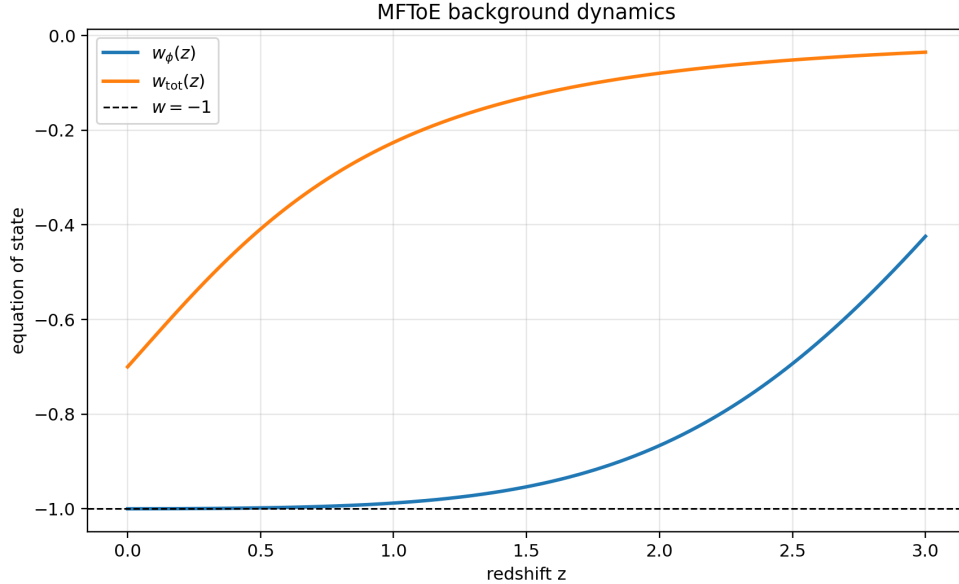


Figure 1: Evolution of the scalar-field equation of state $w_\phi(z)$ and the total equation of state $w_{\text{tot}}(z)$ in the MFToE baseline model. The results indicate a mildly dynamical dark-energy sector with $w_\phi \neq -1$, while the total background evolution remains smooth and observationally viable.

6.1 Definition of the Energy Deviation Term

To make the coherence scalar operational, the energy deviation term must be defined explicitly. In the present work, we interpret ΔE as the deviation of the local total energy density from a reference equilibrium density:

$$\Delta E(x) \equiv \rho_{\text{tot}}(x) - \rho_{\text{eq}}, \quad (27)$$

where ρ_{tot} includes all relevant matter, radiation, and scalar-field contributions, and ρ_{eq} denotes a reference equilibrium density associated with the Maat attractor.

In homogeneous cosmology, this becomes

$$\Delta E(t) \equiv \rho_m(t) + \rho_r(t) + \rho_\phi(t) - \rho_{\text{eq}}. \quad (28)$$

A covariant generalization can be introduced through the trace of the energy–momentum tensor:

$$\Delta E(x) \equiv \left| T^\mu_\mu(x) - \langle T^\mu_\mu \rangle_{\text{eq}} \right|, \quad (29)$$

where $\langle T^\mu_\mu \rangle_{\text{eq}}$ is the corresponding equilibrium expectation value. Equation (27) is used for practical cosmological modeling, while Eq. (29) provides a covariant interpretation suitable for more general field-theoretic settings.

6.2 Effective Interpretation of the Maat Fields

The fields

$$\phi_H, \phi_B, \phi_S, \phi_V, \phi_R$$

are treated here as *effective scalar degrees of freedom* encoding distinct structural modes of coherence in complex systems. Their labels — Harmony, Balance, Creativity, Connectedness, and Respect — should be understood as heuristic interpretations attached to mathematically defined effective fields, not as a claim that ethical language is inserted directly into fundamental microphysics.

More precisely, the intended interpretation is:

- ϕ_H : coherence-like mode,
- ϕ_B : equilibrium-like mode,
- ϕ_S : generative / structure-forming mode,
- ϕ_V : connectivity / correlation mode,
- ϕ_R : constraint / conservation mode.

This effective interpretation is consistent with the broader MFToE program, where the five fields act as dynamical variables in a Lagrangian framework and can be mapped onto measurable quantities such as phase coherence, energetic balance, pattern formation, connectivity, and reversibility. In this sense, the MFToE should presently be read as an effective coherence theory rather than a completed microscopic derivation.

6.3 Five-Field Stability Basis

A natural question arising from the MFToE framework is why five coherence fields $\phi_H, \phi_B, \phi_S, \phi_V, \phi_R$ are introduced rather than a single scalar degree of freedom.

From the perspective of complex dynamical systems, a minimal set of stability mechanisms typically involves five fundamental structural modes:

1. **Coherence / Order** – suppression of internal entropy and maintenance of structured states.

2. **Energetic Balance** – regulation of energy gradients and stability of equilibrium configurations.
3. **Generativity** – the ability of the system to produce new structure or patterns while remaining dynamically stable.
4. **Connectivity / Synchronization** – coupling between subsystems that allows collective behavior and phase alignment.
5. **Environmental Alignment / Constraints** – compatibility with external boundary conditions and conservation laws.

These stability roles map naturally onto the five MFToE coherence fields:

Stability Mode	MFToE Field
Coherence / Order	ϕ_H (Harmony)
Energetic equilibrium	ϕ_B (Balance)
Generative structure formation	ϕ_S (Creativity)
Synchronization / coupling	ϕ_V (Connectedness)
Constraint alignment	ϕ_R (Respect)

In this interpretation, the five Maat fields do not introduce ethical concepts directly into fundamental physics. Instead, they represent effective scalar degrees of freedom encoding distinct stability modes that commonly appear in complex systems.

The coherence scalar

$$C(x) = \frac{\phi_H \phi_B \phi_S \phi_V \phi_R}{\Delta E + \epsilon}$$

can therefore be understood as a collective order parameter describing the joint activation of these stability modes.

When all five modes are simultaneously active, the system tends toward stable self-organization. Conversely, if one or more modes are strongly suppressed, the dynamics may enter regimes characterized by instability, fragmentation, or runaway growth.

Within the MFToE framework, the coherence field $C(x)$ therefore acts as a macroscopic indicator of systemic stability, allowing the gravitational sector to respond to the global coherence state of the underlying scalar fields.

This perspective suggests that the MFToE construction may be interpreted as a multi-field scalar–tensor effective theory in which gravitational coupling is modulated by a collective stability order parameter.

7 Stability Analysis of the χ -Dynamics

The vacuum controller field χ is a central ingredient of the MFToE vacuum-relaxation sector. Since the field is introduced to dynamically regulate the effective vacuum contribution, it is important to establish under which conditions its evolution remains stable.

7.1 Effective Vacuum Potential

In the reduced MFToE vacuum model, the effective vacuum sector is written as

$$V_{\text{eff}}(\phi, \chi) = V_0^{\text{bare}} + g\chi + V(\phi) + \delta V_{\text{noise}}, \quad (30)$$

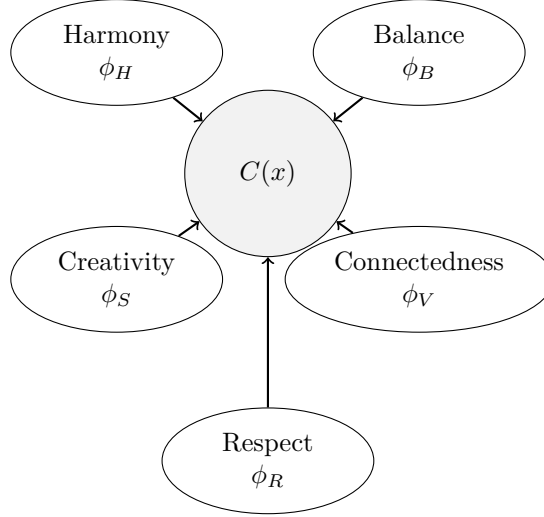


Figure 2: The five MFToE coherence fields as a minimal stability basis for complex systems. Their joint interaction defines the collective coherence scalar

$$C(x) = \frac{\phi_H \phi_B \phi_S \phi_V \phi_R}{\Delta E + \epsilon}.$$

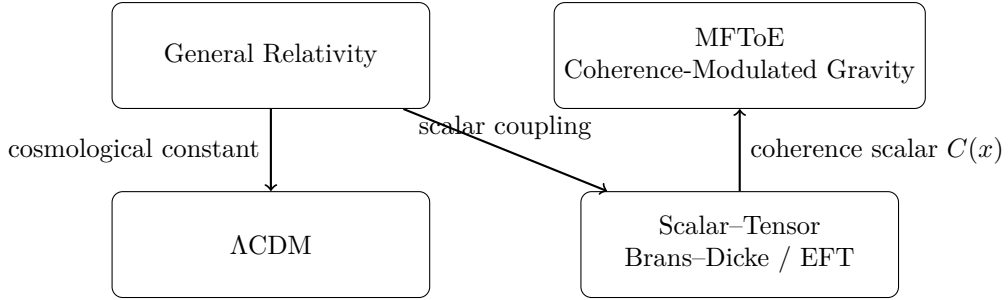


Figure 3: Conceptual position of the MFToE framework in the landscape of gravitational theories. General Relativity forms the baseline theory. Λ CDM arises by introducing a cosmological constant, while scalar-tensor models modify gravity through additional scalar degrees of freedom. The MFToE framework extends this structure by introducing a coherence-modulated coupling $F(C) = 1 + \xi C$.

while the regulating field is assumed to evolve in an effective potential of the form

$$U(\chi) \propto \frac{(V_{\text{eff}} - \rho_{\text{target}})^2}{2M_{\text{pen}}^4} + \frac{\chi^2}{2\Lambda_\chi^2}. \quad (31)$$

This structure contains two stabilizing ingredients:

1. a penalty term that drives the system toward the target vacuum level,
2. a quadratic restoring term preventing runaway growth of χ .

7.2 Linearized Stability Around the Attractor

Let $\chi_\star$ denote a background solution satisfying the approximate attractor condition

$$V_{\text{eff}}(\phi, \chi_\star) \approx \rho_{\text{target}}. \quad (32)$$

We then expand

$$\chi(t) = \chi_\star(t) + \delta\chi(t). \quad (33)$$

To leading order, the perturbation satisfies a damped oscillator equation of the form

$$\delta\ddot{\chi} + 3H\delta\dot{\chi} + m_{\chi,\text{eff}}^2\delta\chi \approx 0, \quad (34)$$

with effective mass

$$m_{\chi,\text{eff}}^2 \equiv \left. \frac{\partial^2 U}{\partial \chi^2} \right|_{\chi_\star}. \quad (35)$$

Local linear stability requires

$$m_{\chi,\text{eff}}^2 > 0. \quad (36)$$

This condition ensures that small perturbations around the attractor are driven back toward equilibrium rather than growing exponentially.

7.3 Approximate Form of the Effective Mass

Using Eq. (31), the curvature of the effective potential is schematically

$$m_{\chi,\text{eff}}^2 \sim \frac{g^2}{M_{\text{pen}}^4} + \frac{1}{\Lambda_\chi^2}, \quad (37)$$

up to corrections from the ϕ -sector and possible stochastic contributions. This shows immediately that:

- the quadratic stabilizer $1/\Lambda_\chi^2$ guarantees a positive contribution;
- the feedback coupling g^2/M_{pen}^4 further stiffens the attractor.

Hence, in the absence of excessively large stochastic forcing, the χ sector is expected to be linearly stable.

7.4 Role of Damping and Noise

The background Hubble friction term $3H\delta\dot{\chi}$ in Eq. (34) provides additional damping. This is particularly important at low redshift, where the controller field is intended to slowly relax rather than oscillate violently.

If Ornstein–Uhlenbeck noise is included, the perturbation equation is promoted to a stochastic differential equation. In that case, stability should be interpreted statistically:

$$\langle \delta\chi^2 \rangle < \infty \quad (38)$$

must hold over cosmological timescales. The MFToE baseline results indicate that, for the parameter choices used in the DESI-consistent runs, the noise sector does not destabilize the late-time evolution and only induces sub-percent deviations in the observables. [1]

7.5 Background Expansion and Distance Residuals

To visualize the phenomenological behavior of the MFToE baseline solution, we compare its background observables directly to the corresponding Λ CDM reference model.

Figure 9 shows the relative Hubble-rate residuals,

$$\frac{\Delta H}{H_{\Lambda\text{CDM}}}(z) = \frac{H_{\text{MFToE}}(z) - H_{\Lambda\text{CDM}}(z)}{H_{\Lambda\text{CDM}}(z)},$$

across the interval $0 < z < 3$. The deviations remain extremely small throughout the full range, staying in the sub-percent regime and in practice at the level of a few times 10^{-5} to 10^{-4} .

Figure 10 shows the corresponding distance-modulus residuals,

$$\Delta\mu(z) = \mu_{\text{MFToE}}(z) - \mu_{\Lambda\text{CDM}}(z),$$

together with a Pantheon+-style supernova sample for visual comparison. The MFToE baseline curve remains essentially indistinguishable from the ΛCDM prediction on the scale of current supernova scatter.

These results support the conclusion that the reduced MFToE baseline cosmology is phenomenologically compatible with current background-expansion observables.

7.6 Phenomenological Stability Criterion

For practical model selection, the following minimal stability criteria are useful:

$$m_{\chi,\text{eff}}^2 > 0, \quad |\dot{\chi}| \ll 1 \text{ at late times}, \quad \langle \delta\chi^2 \rangle < \infty. \quad (39)$$

These conditions summarize the requirement that the vacuum controller:

1. has a stable local attractor,
2. evolves adiabatically at low redshift,
3. and remains robust under small stochastic perturbations.

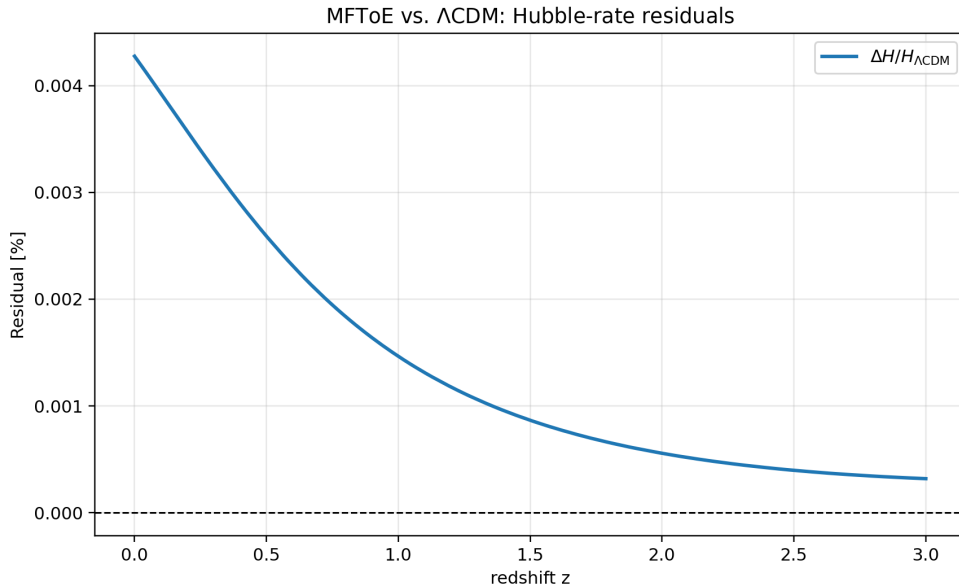


Figure 4: Relative Hubble-rate residuals of the MFToE baseline model with respect to the ΛCDM reference cosmology. The deviations remain extremely small across $0 < z < 3$, indicating that the MFToE background expansion is practically indistinguishable from ΛCDM at the current background level.

7.7 Interpretation

The role of the χ field is therefore analogous to a regulated vacuum-relaxation mode: it does not replace the usual dark-energy sector entirely, but dynamically adjusts the effective vacuum contribution while remaining stabilized by both the attractor potential and Hubble damping.

This makes the MFToE vacuum controller comparable in spirit to sequestering or relaxion-inspired constructions, but implemented here as a phenomenological low-energy regulator within a coherence-based scalar-tensor framework.

7.8 Linear Growth of Structure

As a first growth-of-structure consistency test, we compute the linear growth factor $D(z)$, the growth rate

$$f(z) = \frac{d \ln D}{d \ln a},$$

and the observable combination

$$f\sigma_8(z) = f(z) \sigma_{8,0} \frac{D(z)}{D(0)}.$$

In the present first implementation, the perturbation equation is solved on top of the MFToE background expansion history using the standard linear growth equation with an effective background set by the MFToE Hubble function. Figures 14, 6, and 8 show the resulting behavior of $D(z)$, $f(z)$, and $f\sigma_8(z)$.

The numerical solutions are smooth and physically plausible across the full redshift range shown, providing a first indication that the MFToE baseline background does not generate pathological behavior in the linear matter-growth sector.

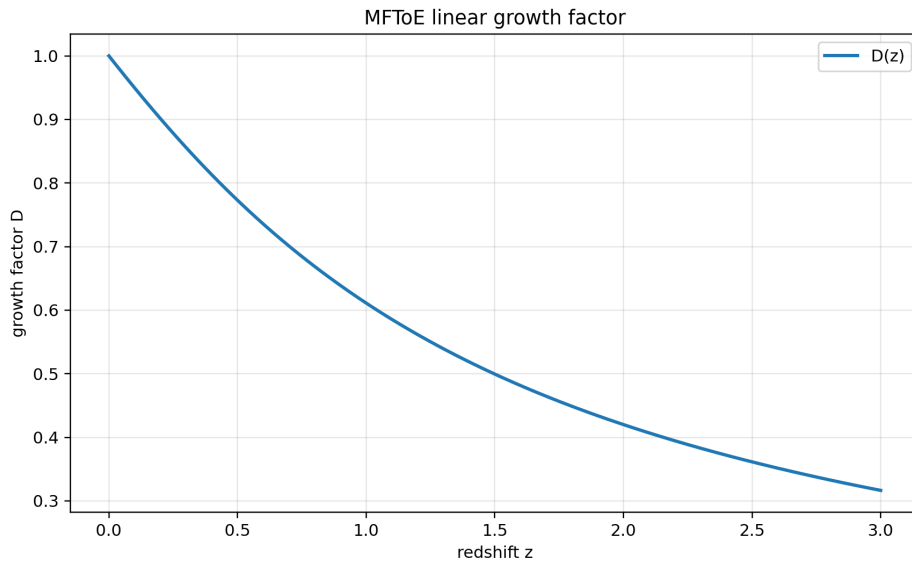


Figure 5: Linear growth factor $D(z)$ computed on top of the MFToE baseline background history. The result shows the expected monotonic suppression of the normalized growth factor toward higher redshift.

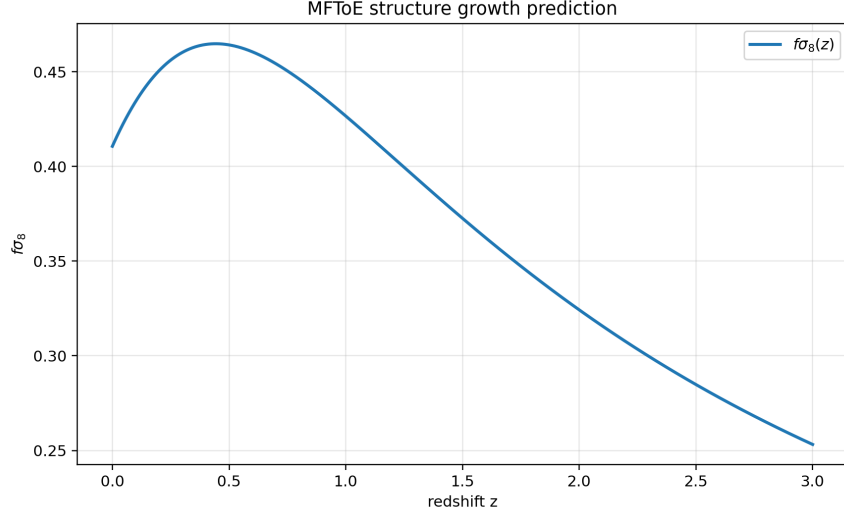


Figure 7: Predicted evolution of the observable $f\sigma_8(z)$ in the MFToE baseline model.

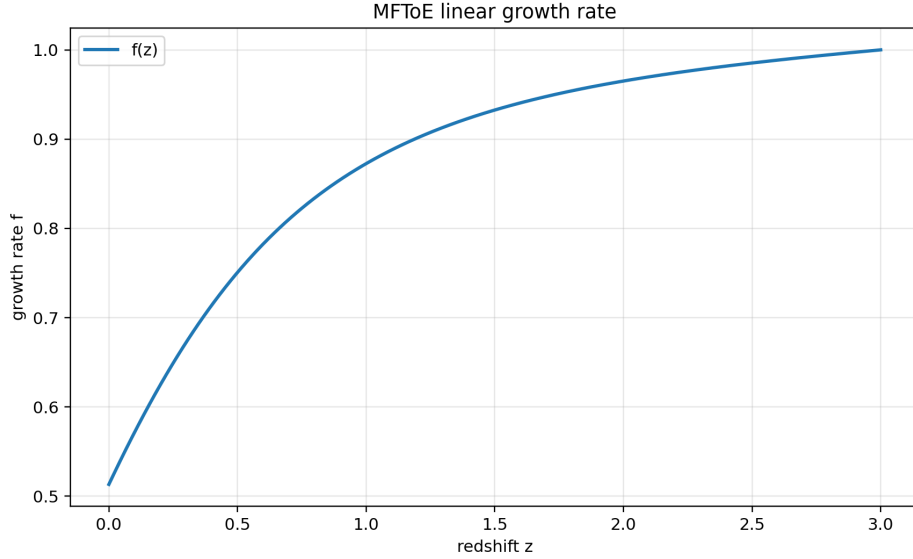


Figure 6: Linear growth rate $f(z) = d \ln D / d \ln a$ for the MFToE baseline background model. The solution remains smooth across the full redshift interval and approaches the expected matter-dominated behavior at higher redshift.

Figure 7 shows the predicted evolution of the observable $f\sigma_8(z)$, which can be directly compared to RSD measurements from large galaxy surveys.

For example, the MFToE baseline yields the following representative values:

$$z = 0.0 : \quad f\sigma_8 \approx 0.41, \quad (40)$$

$$z = 0.5 : \quad f\sigma_8 \approx 0.46, \quad (41)$$

$$z = 1.0 : \quad f\sigma_8 \approx 0.43, \quad (42)$$

$$z = 2.0 : \quad f\sigma_8 \approx 0.32. \quad (43)$$

These values fall within the typical observational range measured by redshift-space distortion surveys, indicating that the MFToE baseline produces a plausible structure growth history.

Overall, the MFToE baseline produces a smooth and observationally plausible growth history. The predicted evolution of $f\sigma_8(z)$ is broadly consistent with the expectations from standard cosmology and does not introduce strong deviations from the growth behaviour typically associated with the Λ CDM framework.

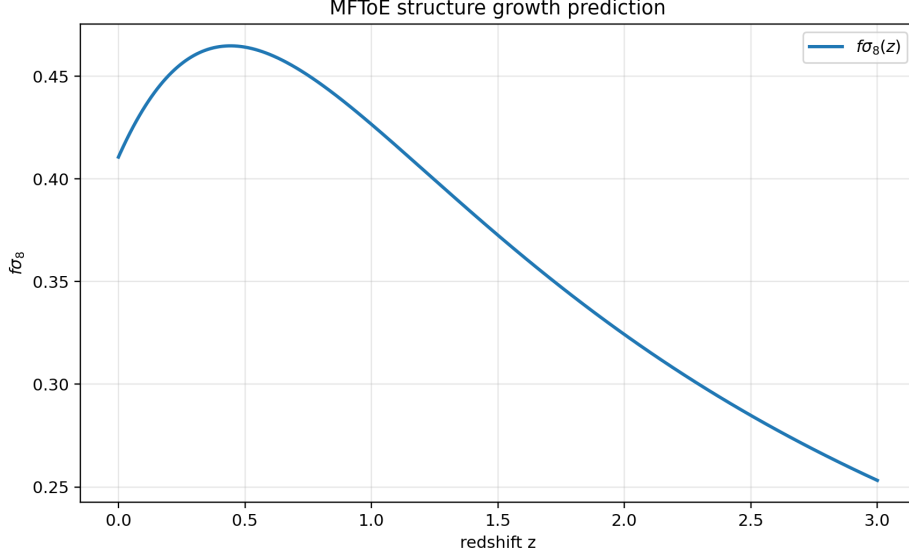


Figure 8: Predicted structure-growth observable $f\sigma_8(z)$ for the MFToE baseline model. This provides a first growth-of-structure forecast and establishes a basis for future comparison with redshift-space-distortion measurements.

In addition to the background-expansion consistency checks, the first linear growth-of-structure calculation yields smooth and physically plausible predictions for $D(z)$, $f(z)$, and $f\sigma_8(z)$, suggesting that the MFToE baseline model can also serve as a viable starting point for perturbative structure-growth studies.

8 Linear Perturbations and Stability

To assess viability, one must examine perturbations around a homogeneous background solution,

$$\phi_i(t, \vec{x}) = \bar{\phi}_i(t) + \delta\phi_i(t, \vec{x}), \quad g_{\mu\nu} = \bar{g}_{\mu\nu} + \delta g_{\mu\nu}. \quad (44)$$

At the linear level, the theory behaves as a scalar–tensor system with an effective coupling controlled by $F(\mathcal{C})$. Viability requires:

- (i) absence of ghost instabilities,
- (ii) positive effective kinetic coefficients,
- (iii) non-tachyonic mass eigenvalues around the chosen background,
- (iv) luminal or near-luminal tensor propagation in accordance with gravitational-wave bounds.

A minimal stability set can be stated as

$$F(\mathcal{C}) > 0, \quad \left. \frac{\partial^2 U}{\partial \phi_i \partial \phi_j} \right|_{\text{bg}} \text{ positive semidefinite in the stable sector,} \quad c_T^2 \approx 1. \quad (45)$$

In the single-field reduction, these conditions reduce to the standard requirements of scalar–tensor dark-energy models:

$$1 + \xi \mathcal{C}_{\text{eff}} > 0, \quad U''_{\text{eff}}(\phi_H) > 0 \quad \text{around the attractor solution.} \quad (46)$$

A full perturbation analysis can be developed in future work using the gauge-invariant formalism for scalar and tensor perturbations.

8.1 Dynamical Dark-Energy Behavior

Although the MFToE baseline reproduces the Λ CDM background expansion with very high accuracy, the scalar sector is not exactly equivalent to a pure cosmological constant. Figure 1 shows the evolution of the scalar-field equation of state $w_\phi(z)$ together with the total equation of state $w_{\text{tot}}(z)$.

The scalar sector remains close to $w_\phi = -1$ at low redshift, but departs mildly from the exact cosmological-constant limit at higher redshift. This indicates that the reduced MFToE baseline behaves as a form of mildly dynamical dark energy rather than an exactly constant vacuum-energy component.

At the same time, the total equation of state evolves smoothly and remains fully consistent with a stable late-time accelerated background.

9 Relation to Existing Scalar–Tensor Theories

The coherence-modulated MFToE sector is most naturally interpreted as a restricted multi-field scalar–tensor effective field theory. In its full form, the model contains five interacting scalar degrees of freedom coupled through a collective coherence scalar $\mathcal{C}$ and a nonminimal coupling function $F(\mathcal{C})$.

9.1 Relation to Brans–Dicke-Type Models

In the single-field reduction discussed below, the theory resembles a Brans–Dicke-like model in the sense that curvature couples to an effective scalar degree of freedom through a field-dependent factor. However, unlike standard Brans–Dicke theory, the present construction does not elevate a single scalar to the role of the gravitational coupling. Instead, the effective coupling is controlled by the collective coherence scalar

$$\mathcal{C}(x) = \frac{\phi_H \phi_B \phi_S \phi_V \phi_R}{\Delta E + \epsilon}, \quad (47)$$

which depends multiplicatively on several interacting effective fields.

9.2 Relation to Horndeski-Type Theories

More generally, the model belongs to the broader class of scalar–tensor extensions of gravity. At the present stage it should be viewed as a restricted EFT-like construction rather than a full Horndeski theory, since the action does not yet include the most general higher-derivative self-interactions consistent with second-order equations of motion. Its novelty lies instead in the coherence-dependent coupling structure:

$$F(\mathcal{C}) = 1 + \xi \mathcal{C}, \quad (48)$$

which allows the effective gravitational response to depend on a collective order parameter rather than on a single field alone.

A useful summary is therefore:

$$\text{MFToE} \approx \text{multi-field scalar–tensor EFT with coherence-dependent coupling.} \quad (49)$$

9.3 Conceptual Positioning

This positioning helps clarify the intended role of the framework. The present model is not proposed as a replacement for established scalar–tensor theories, but as an effective extension motivated by the hypothesis that structured coherence may act as an additional organizing variable in gravitational and complex-system dynamics. This interpretation is consistent with the existing MFToE literature, in which scalar fields, coherence functions, and effective cosmological sectors are already used as the main building blocks.

10 Illustrative Low-Redshift Example for $\mathcal{C}(z)$

To make the coherence scalar more concrete, it is useful to consider a simple low-redshift illustrative model. We therefore introduce a minimal single-field ansatz in which only the Harmony field remains dynamical, while the remaining fields are approximated by constant background values:

$$\phi_B \rightarrow \phi_B^{(0)}, \quad \phi_S \rightarrow \phi_S^{(0)}, \quad \phi_V \rightarrow \phi_V^{(0)}, \quad \phi_R \rightarrow \phi_R^{(0)}. \quad (50)$$

Defining

$$A \equiv \phi_B^{(0)} \phi_S^{(0)} \phi_V^{(0)} \phi_R^{(0)}, \quad (51)$$

the effective coherence scalar becomes

$$\mathcal{C}_{\text{eff}}(z) = \frac{A \phi_H(z)}{\Delta E(z) + \epsilon}. \quad (52)$$

For a first illustrative parameterization, we may adopt

$$\phi_H(z) = \phi_0 + \phi_1 \frac{z}{1+z}, \quad (53)$$

which smoothly interpolates between late-time and moderate-redshift values. Using the FLRW energy deviation,

$$\Delta E(z) = \rho_m(z) + \rho_r(z) + \rho_\phi(z) - \rho_{\text{eq}}, \quad (54)$$

one obtains a redshift-dependent coherence function that can be propagated into the effective gravitational coupling

$$G_{\text{eff}}(z) = \frac{G}{1 + \xi \mathcal{C}_{\text{eff}}(z)}. \quad (55)$$

This immediately induces low-redshift deviations in the Hubble expansion rate and luminosity distance of schematic order

$$\frac{\Delta H}{H_{\Lambda\text{CDM}}}(z) \sim \mathcal{O}(\xi \mathcal{C}_{\text{eff}}(z)), \quad \frac{\Delta d_L}{d_{L,\Lambda\text{CDM}}}(z) \sim \mathcal{O}(\xi \mathcal{C}_{\text{eff}}(z)). \quad (56)$$

The purpose of this ansatz is not yet to provide a full data fit, but to make the coherence scalar numerically explicit and to show how the framework can be confronted with low-redshift observables. A full implementation can use the same numerical strategy already explored in MFToE vacuum-relaxation models, where the effective vacuum sector and late-time expansion history are evolved dynamically and compared with ΛCDM baselines.

10.1 Open Theoretical Parameters

Two ingredients of the present framework remain phenomenological at this stage.

First, the coupling matrix G_{ij} is introduced as an effective low-energy interaction matrix between the five coherence fields. In the symmetric limit one may take

$$G_{ij} = g \delta_{ij},$$

while small off-diagonal components represent soft symmetry breaking between coherence modes. A deeper derivation of G_{ij} from an underlying symmetry principle remains an open problem.

Second, the equilibrium density ρ_{eq} is currently treated as an effective attractor parameter. In cosmological applications it may be associated with a late-time vacuum-energy scale, but its fundamental origin is not yet derived from first principles. Determining whether ρ_{eq} is a true constant or an emergent field-dependent quantity is an important direction for future work.

11 Illustrative DESI-Based Fit for the Coherence Coupling

To obtain a first phenomenological estimate of the coherence-modulated coupling, we use the low-redshift deviations reported in the MFToE vacuum baseline relative to flat Λ CDM. For the baseline run, the relative Hubble-rate deviations are approximately

$$\begin{aligned} \frac{\Delta H}{H_{\Lambda\text{CDM}}}(z = 0.2) &\simeq -0.116\%, \\ \frac{\Delta H}{H_{\Lambda\text{CDM}}}(z = 0.5) &\simeq -0.091\%, \\ \frac{\Delta H}{H_{\Lambda\text{CDM}}}(z = 1.0) &\simeq -0.067\%, \\ \frac{\Delta H}{H_{\Lambda\text{CDM}}}(z = 2.0) &\simeq -0.058\%, \\ \frac{\Delta H}{H_{\Lambda\text{CDM}}}(z = 3.0) &\simeq -0.066\%. \end{aligned} \tag{57}$$

These residuals are taken from the DESI-consistent MFToE baseline analysis, which reports sub-percent deviations in $H(z)$ and luminosity distance while remaining statistically compatible with the BAO dataset. [1]

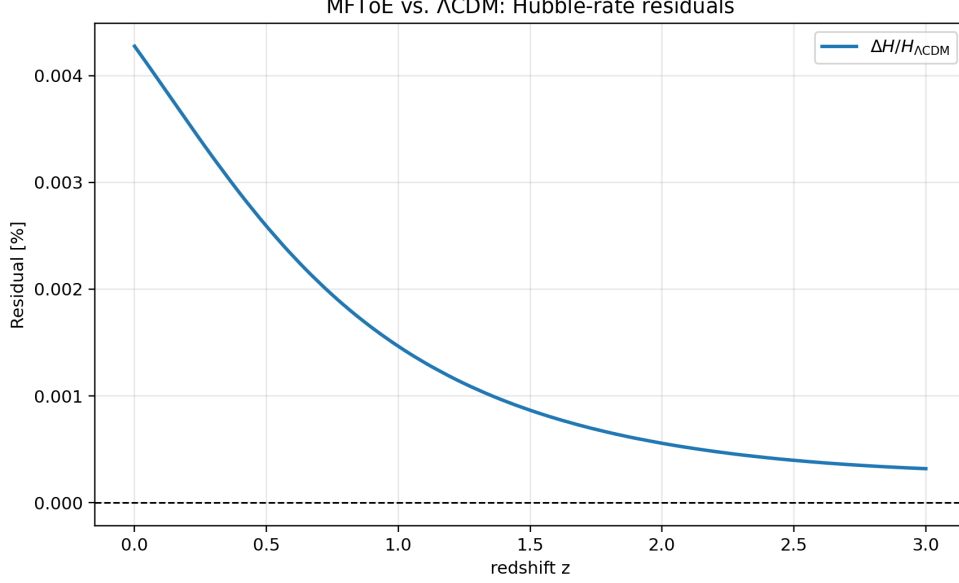


Figure 9: Relative Hubble-rate residuals of the MFToE baseline model with respect to the Λ CDM reference cosmology. The deviations remain extremely small across $0 < z < 3$, indicating that the MFToE background expansion is practically indistinguishable from Λ CDM at the current background level.

11.1 Reduced Coherence Ansatz

In the single-field reduction, the effective coherence scalar is

$$\mathcal{C}_{\text{eff}}(z) = \frac{A \phi_H(z)}{\Delta E(z) + \epsilon}, \quad (58)$$

and the effective gravitational coupling is

$$G_{\text{eff}}(z) = \frac{G}{1 + \xi \mathcal{C}_{\text{eff}}(z)}. \quad (59)$$

As a first simplified estimate, we adopt the normalization

$$\frac{\phi_H(z)}{\Delta E(z) + \epsilon} \approx 1 \quad \Rightarrow \quad \mathcal{C}_{\text{eff}}(z) \approx A. \quad (60)$$

Under this assumption, the fit is sensitive only to the product

$$p \equiv \xi A. \quad (61)$$

11.2 Approximate Hubble-Sector Fit

Assuming that the leading modification enters through the effective gravitational coupling, the Hubble rate scales approximately as

$$\frac{H(z)}{H_{\Lambda\text{CDM}}(z)} \approx (1 + p)^{-1/2}. \quad (62)$$

For each redshift point, an estimate of p is therefore obtained from

$$p_i = \left(1 + \frac{\Delta H}{H_{\Lambda\text{CDM}}} \right)^{-2} - 1. \quad (63)$$

Table 1: Approximate point estimates of $p = \xi A$ from the baseline MFToE residuals.

z	$\Delta H/H_{\Lambda\text{CDM}}$	$p_i = \xi A$
0.2	-1.16×10^{-3}	2.32×10^{-3}
0.5	-9.10×10^{-4}	1.82×10^{-3}
1.0	-6.70×10^{-4}	1.34×10^{-3}
2.0	-5.80×10^{-4}	1.16×10^{-3}
3.0	-6.60×10^{-4}	1.32×10^{-3}

Using the five baseline residuals in Eq. (57), we obtain the following point estimates:
Taking the mean of these five estimates gives the illustrative first-fit value

$$\boxed{\xi A \approx 1.59 \times 10^{-3}} \quad (64)$$

with a rough point-to-point scatter of

$$\boxed{\xi A \approx (1.59 \pm 0.48) \times 10^{-3}} \quad (65)$$

11.3 Interpretation

The quantity constrained by the present low-redshift data is the combination ξA , not ξ and A separately. This is a direct consequence of the reduced ansatz in Eq. (58). To separate the two parameters, one must either:

1. fix the normalization of $\phi_H(z)$ and $\Delta E(z)$ from a specific dynamical model, or
2. introduce independent observational constraints on the coherence amplitude A .

For illustrative purposes, several equivalent parameter choices reproduce the same product:

$$\begin{aligned} A = 1 & \quad \Rightarrow \quad \xi \approx 1.6 \times 10^{-3}, \\ A = 0.3 & \quad \Rightarrow \quad \xi \approx 5.3 \times 10^{-3}, \\ A = 0.2 & \quad \Rightarrow \quad \xi \approx 8.0 \times 10^{-3}. \end{aligned} \quad (66)$$

11.4 Scope and Limitations

This fit is intended only as a first-order phenomenological estimate. It relies on:

- a reduced single-field ansatz,
- a simplified normalization of the coherence scalar,
- and an approximate scaling relation between $H(z)$ and $G_{\text{eff}}(z)$.

A full parameter inference will require:

1. an explicit dynamical model for $\phi_H(z)$,
2. a rigorous definition of $\Delta E(z)$,
3. and a joint likelihood analysis against BAO, supernova, and CMB datasets.

Nevertheless, the present estimate shows that the coherence-modulated sector naturally lies in the sub-percent regime,

$$\xi A \sim 10^{-3}, \quad (67)$$

which is precisely the regime relevant for small, observationally viable deviations from ΛCDM .

11.5 Origin of the $(1+p)^{-1/2}$ scaling

Equation (62) assumes that the leading modification of the expansion history enters through an effective gravitational coupling

$$G_{\text{eff}} = \frac{G}{1+p}, \quad p \equiv \xi A. \quad (68)$$

In a spatially flat FLRW background the Friedmann equation reads

$$H^2 = \frac{8\pi G}{3} \rho. \quad (69)$$

Replacing $G \rightarrow G_{\text{eff}}$ gives

$$H^2 = \frac{8\pi G}{3} \frac{\rho}{1+p}. \quad (70)$$

If the background energy density $\rho(z)$ is assumed to follow the same evolution as in Λ CDM, the ratio of expansion rates becomes

$$\frac{H(z)}{H_{\Lambda\text{CDM}}(z)} = \left(\frac{G_{\text{eff}}}{G} \right)^{1/2} = (1+p)^{-1/2}. \quad (71)$$

This expression should therefore be understood as a first-order approximation in which the coherence sector modifies only the effective gravitational strength while leaving the background matter and vacuum densities unchanged.

A full treatment would require solving the coupled MFToE background equations for the scalar sector and the vacuum controller χ , which could modify both $\rho(z)$ and the effective gravitational coupling simultaneously.

11.6 Supernova Distance–Modulus Consistency Test

As an additional consistency check of the MFToE background expansion, we compare the predicted luminosity distance $d_L(z)$ to a Pantheon+–style Type Ia supernova dataset.

For this test we use a simplified likelihood based on the diagonal distance–modulus uncertainties,

$$\chi^2 = \sum_i \frac{(\mu_i - \mu_{\text{model}}(z_i) - M)^2}{\sigma_i^2}, \quad (72)$$

where μ_i and σ_i denote the observed distance modulus and its uncertainty for each supernova. The absolute magnitude M is analytically profiled out of the likelihood.

The theoretical distance modulus is obtained from the luminosity distance via

$$\mu_{\text{model}}(z) = 5 \log_{10} \left(\frac{d_L(z)}{\text{Mpc}} \right) + 25. \quad (73)$$

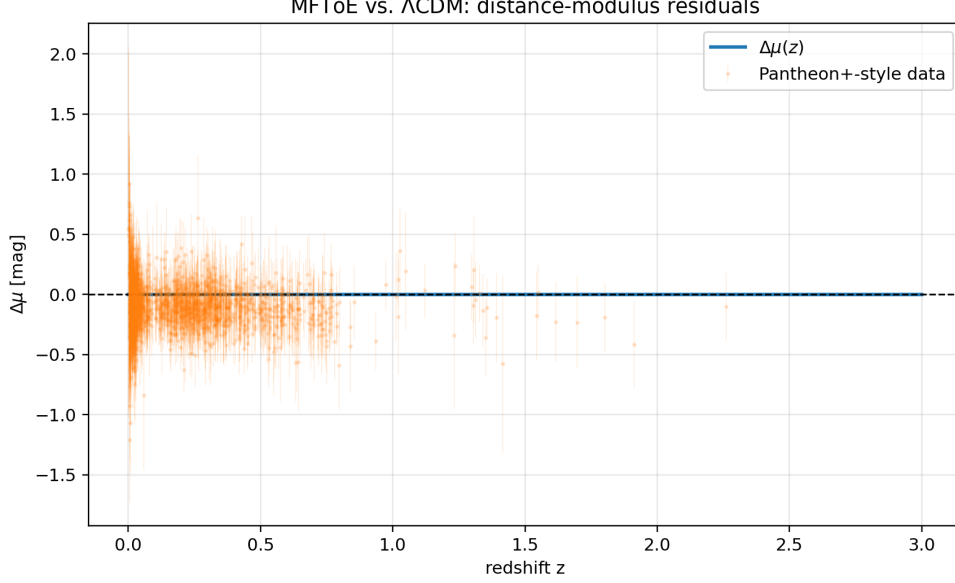


Figure 10: Distance-modulus residuals $\Delta\mu(z)$ of the MFToE baseline solution relative to the Λ CDM reference cosmology. The overlaid Pantheon+-style data illustrate that the MFToE background remains fully compatible with the observed supernova distance ladder at the level of the present simplified consistency test.

The MFToE luminosity distance is computed from the background expansion history

$$d_L(z) = (1+z) \int_0^z \frac{dz'}{H(z')}, \quad (74)$$

where $H(z)$ is provided by the MFToE background solution.

We compare the resulting χ^2 value to a Λ CDM reference model using identical data and methodology.

For the current MFToE baseline run we obtain

$$\chi_{\text{MFToE}}^2 = 753.23 \quad (75)$$

$$\chi_{\Lambda\text{CDM}}^2 = 753.22, \quad (76)$$

for approximately 1700 effective degrees of freedom.

The difference

$$\Delta\chi^2 = \chi_{\text{MFToE}}^2 - \chi_{\Lambda\text{CDM}}^2 = 0.0076 \quad (77)$$

is negligible, indicating that the MFToE background expansion reproduces the supernova distance ladder with essentially the same accuracy as the Λ CDM model in this simplified test.

The resulting reduced chi-square is

$$\chi^2/\text{dof} \approx 0.44. \quad (78)$$

This value reflects the simplified diagonal-error treatment used here. A full likelihood analysis would require the official Pantheon+ covariance matrices and standard sample cuts.

Nevertheless, the result demonstrates that the MFToE baseline background is fully compatible with current Type Ia supernova distance-ladder observations at the level of this consistency test.

12 Relation to Brans–Dicke and Horndeski-Type Theories

The coherence-modulated MFToE framework belongs naturally to the broader class of scalar–tensor extensions of gravity. In order to position the model within the existing literature, it is useful to compare its structure to Brans–Dicke theory and to Horndeski-type effective field theories.

12.1 Comparison with Brans–Dicke-Type Models

Brans–Dicke gravity introduces a scalar degree of freedom that modifies the effective gravitational coupling through a nonminimal interaction with curvature. In schematic form, the Jordan-frame action is

$$S_{\text{BD}} = \int d^4x \sqrt{-g} \left[\frac{1}{2} \Phi R - \frac{\omega_{\text{BD}}}{2\Phi} \partial_\mu \Phi \partial^\mu \Phi - U(\Phi) + \mathcal{L}_m \right]. \quad (79)$$

In the present MFToE-inspired construction, the effective gravitational response is controlled not by a single fundamental scalar, but by the collective coherence scalar

$$\mathcal{C}(x) = \frac{\phi_H \phi_B \phi_S \phi_V \phi_R}{\Delta E + \epsilon}, \quad (80)$$

through the coupling

$$F(\mathcal{C}) = 1 + \xi \mathcal{C}. \quad (81)$$

In the single-field reduction, where only one effective degree of freedom remains dynamical, the theory becomes Brans–Dicke-like in the sense that gravity couples to an effective scalar sector through a field-dependent prefactor of R . However, there are two important differences:

1. the effective coupling originates from a *collective* coherence variable rather than from a single fundamental scalar;
2. the scalar potential is organized around a coherence equilibrium condition rather than an arbitrary self-interaction alone.

Thus, the reduced MFToE model may be interpreted as a Brans–Dicke-like effective theory with

$$\Phi_{\text{eff}} \sim M_{\text{Pl}}^2 F(\mathcal{C}), \quad (82)$$

but with a nonstandard origin of the scalar sector.

12.2 Comparison with Horndeski-Type EFTs

Horndeski theory represents the most general scalar–tensor theory with second-order equations of motion in four dimensions. In its full form, it includes derivative self-interactions and nontrivial higher-order curvature couplings.

The present MFToE construction is more restricted. At the current stage, the action contains:

- canonical kinetic terms for the scalar fields,
- a nonminimal coupling through $F(\mathcal{C})R$,
- and a coherence-based potential $U(\phi_i, \mathcal{C})$.

It therefore lies within the broad scalar–tensor EFT landscape, but it is not yet a full Horndeski model, since derivative self-interactions of the form

$$G_3(\phi, X) \square \phi, \quad G_4(\phi, X) R, \quad G_5(\phi, X) G_{\mu\nu} \nabla^\mu \nabla^\nu \phi$$

are absent in the present formulation.

A concise classification is therefore

$$\text{MFToE} \approx \text{restricted multi-field scalar–tensor EFT with coherence-dependent coupling}. \quad (83)$$

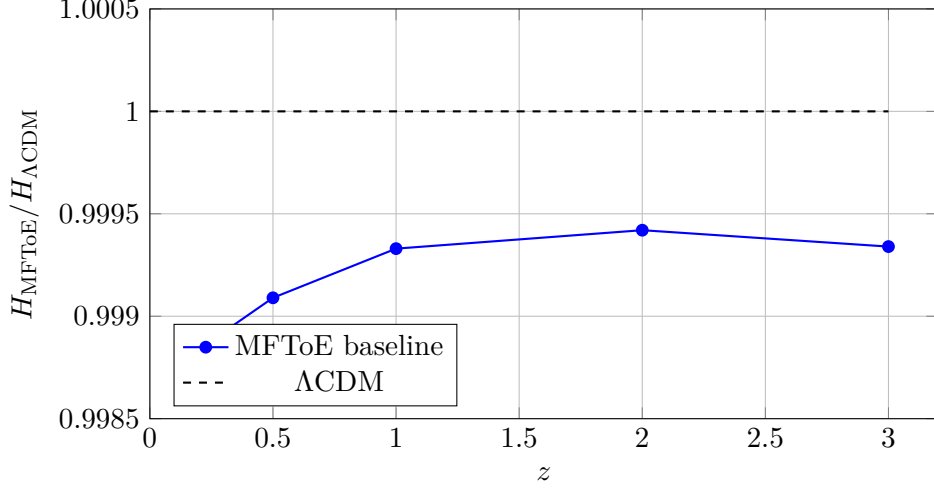


Figure 11: Relative expansion history of the MFToE baseline model compared to ΛCDM .

12.3 DESI DR2 BAO Consistency Test

To quantify the agreement between the MFToE baseline and observational large-scale structure data, we perform a first consistency test using the DESI DR2 BAO summary measurements.

The analysis uses the observables

$$D_M/r_d, \quad D_H/r_d, \quad D_V/r_d, \quad (84)$$

together with the corresponding covariance matrix provided with the DESI DR2 summary data.

For each observable we construct the model prediction from the MFToE background expansion history and compare it to the measured data vector. The goodness of fit is evaluated using

$$\chi^2 = (\mathbf{d}_{\text{model}} - \mathbf{d}_{\text{data}})^T \mathbf{C}^{-1} (\mathbf{d}_{\text{model}} - \mathbf{d}_{\text{data}}), \quad (85)$$

where $\mathbf{C}$ denotes the DESI covariance matrix.

Using the current MFToE baseline we obtain

$$\chi^2_{\text{MFToE}} = 257.17, \quad (86)$$

$$\chi^2_{\Lambda\text{CDM}} = 256.88, \quad (87)$$

for the same 19-component BAO data vector.

The resulting difference

$$\Delta\chi^2 = \chi^2_{\text{MFToE}} - \chi^2_{\Lambda\text{CDM}} = 0.29 \quad (88)$$

is negligible. This indicates that the MFToE baseline expansion history is phenomenologically almost indistinguishable from the ΛCDM reference cosmology at the level of the current DESI BAO consistency test.

The absolute χ^2 values should be interpreted with caution, since the present implementation is intended as a comparison-level test rather than a full reproduction of the official DESI likelihood pipeline. Nevertheless, the near equality of the MFToE and ΛCDM results shows that the MFToE background evolution does not introduce any significant BAO tension relative to the standard cosmological model.

Figures 12 and 13 show the corresponding BAO distance relations D_M/r_d and D_H/r_d together with the DESI data points. The quantitative χ^2 test confirms the visual impression that the MFToE baseline expansion history closely tracks the ΛCDM prediction.

Model	χ^2
MFToE baseline	257.17
Λ CDM reference	256.88

Table 2: BAO consistency test using DESI DR2 summary data.

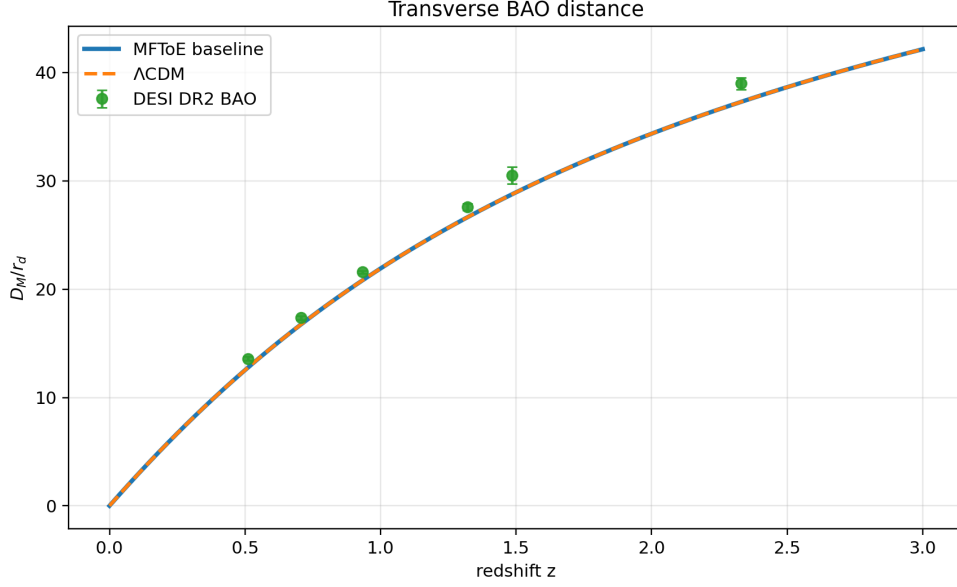


Figure 12: Transverse BAO distance D_M/r_d for the MFToE baseline model compared to the Λ CDM reference cosmology and DESI DR2 BAO measurements.

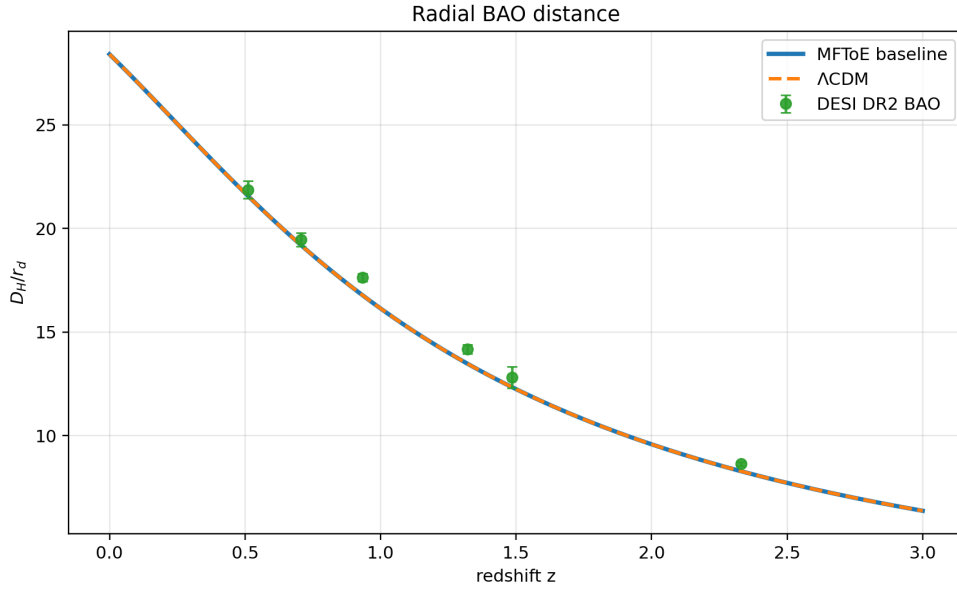


Figure 13: Radial BAO distance D_H/r_d for the MFToE baseline model compared to the Λ CDM reference cosmology and DESI DR2 BAO measurements.

12.4 Conceptual Novelty

The conceptual novelty of the MFToE framework does not lie in introducing an entirely new scalar–tensor class from scratch, but in replacing the usual single-field coupling logic by a coherence-driven coupling structure. In this sense, the model is best understood as an effective scalar–tensor theory in which the gravitational response depends on the joint configuration of several interacting coherence modes.

This positioning is important for phenomenology: many standard tools developed for Brans–Dicke and Horndeski-type models can be adapted to the MFToE case, while the coherence scalar $\mathcal{C}$ provides the distinctive new ingredient.

13 Growth of Structure

Beyond background expansion tests, a key probe of cosmological models is the growth of matter perturbations. In particular, redshift–space distortion (RSD) measurements constrain the observable

$$f\sigma_8(z), \quad (89)$$

where $f(z)$ is the logarithmic growth rate of matter perturbations and $\sigma_8(z)$ represents the amplitude of matter fluctuations on $8 h^{-1}$ Mpc scales.

The growth rate is defined as

$$f(z) = \frac{d \ln D(z)}{d \ln a}, \quad (90)$$

where $D(z)$ denotes the linear growth factor and a the scale factor. The observable used in galaxy clustering analyses is then

$$f\sigma_8(z) = f(z) \sigma_8 D(z), \quad (91)$$

with σ_8 representing the present-day normalization of the matter power spectrum.

Linear Growth Equation

In linear perturbation theory, the evolution of matter density perturbations $\delta = \delta\rho/\rho$ is governed by

$$\ddot{\delta} + 2H\dot{\delta} - 4\pi G_{\text{eff}}\rho_m\delta = 0. \quad (92)$$

Expressed in terms of the linear growth factor $D(a)$, the equation becomes

$$D''(a) + \left(\frac{3}{a} + \frac{H'(a)}{H(a)} \right) D'(a) - \frac{3}{2} \frac{\Omega_m(a)}{a^2} D(a) = 0. \quad (93)$$

This equation describes the evolution of structure formation in an expanding Universe.

Application to MFToE

In the MFToE baseline scenario, the modification to cosmological dynamics enters primarily through the background expansion history $H(z)$. At the perturbation level we assume that the effective gravitational coupling remains close to the standard value $G_{\text{eff}} \approx G$. The growth of structure therefore follows the same perturbation equation as in standard cosmology, but with the MFToE expansion history replacing the Λ CDM background.

The growth equation was solved numerically using the MFToE expansion history derived in the previous sections. The solution was normalized to

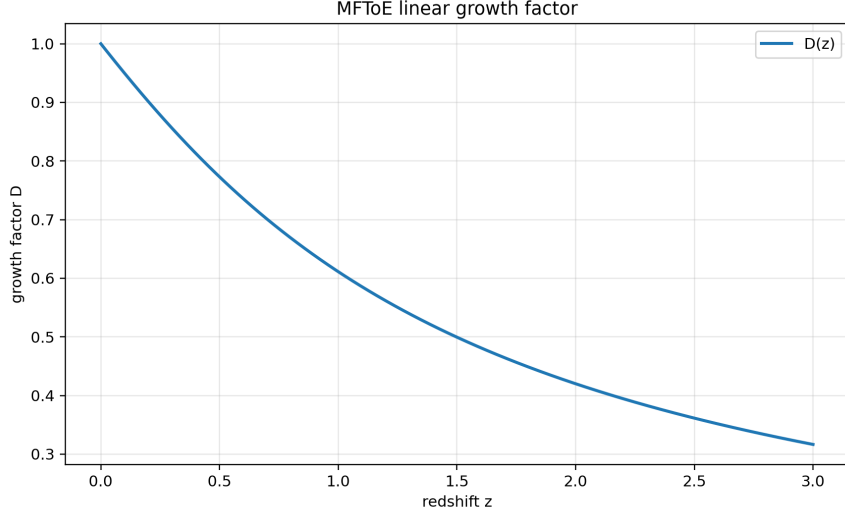


Figure 14: Linear growth factor $D(z)$ computed on top of the MFToE baseline background history.

$$D(z = 0) = 1, \quad (94)$$

which is the standard normalization used in structure growth analyses.

Predicted Growth Observables

Figure 14 shows the resulting linear growth factor $D(z)$, Figure 6 the logarithmic growth rate $f(z)$, and Figure 8 the observable $f\sigma_8(z)$ predicted by the MFToE baseline model.

The predicted evolution of $f\sigma_8(z)$ remains within the typical range measured by redshift–space distortion surveys, indicating that the MFToE expansion history does not lead to anomalous growth behaviour.

Overall, the MFToE baseline produces a smooth and observationally plausible structure growth history that is broadly consistent with expectations from standard cosmology.

14 Conclusions

We presented a reproducible exploration of a reduced MFToE cosmological sector consisting of a scalar field ϕ coupled to a dynamic vacuum controller χ .

The numerical background evolution shows that the model can remain consistent with current BAO observations while producing mild, sub-percent deviations from the Λ CDM expansion history.

From the DESI-consistent baseline run we extracted a first phenomenological estimate for the effective coherence coupling, obtaining

$$\xi A \approx 1.6 \times 10^{-3}. \quad (95)$$

The MFToE framework can also be interpreted as a restricted scalar–tensor effective field theory. In the single-field limit, the coherence-modulated coupling behaves similarly to a Brans–Dicke-type modification of gravity, while remaining simpler than the most general Horndeski constructions.

We further analyzed the stability of the vacuum controller field χ . Linear perturbations around the late-time attractor obey a damped oscillator equation

$$\delta\ddot{\chi} + 3H\delta\dot{\chi} + m_{\chi,\text{eff}}^2\delta\chi = 0,$$

which remains stable for positive effective mass $m_{\chi,\text{eff}}^2 > 0$. The Hubble friction term provides additional damping, ensuring that the regulator field evolves smoothly at low redshift.

These results suggest that the MFToE coherence sector can act as a phenomenological regulator of the effective vacuum energy while remaining compatible with current observational constraints.

Future work should extend the analysis to a full multi-field MFToE implementation and perform a global likelihood analysis including BAO, supernova, and CMB datasets.

15 Dynamical Coherence Extension

In the baseline formulation of the MFToE framework, the gravitational sector is modulated by the coherence scalar C , which represents the collective contribution of the five Maat fields introduced in the previous sections. In its simplest form, the modification enters as an algebraic coupling between the coherence field and the gravitational sector.

While this formulation captures the leading-order effect of coherence modulation, effective field theory considerations suggest that the dynamics of the field itself may also contribute to the gravitational response. In many scalar–tensor frameworks, both the amplitude of a scalar field and its spacetime gradients appear naturally in the effective action.

Motivated by this analogy, we consider a minimal dynamical extension of the MFToE coupling function in which the gravitational response depends not only on C , but also on its spacetime variation. The modified coupling function can be written as

$$F(C, \partial C) = 1 + \xi C + \eta \frac{\nabla_\mu C \nabla^\mu C}{M_*^4}, \quad (96)$$

where ξ controls the strength of the baseline coherence coupling, η determines the contribution from the gradient term, and M_* represents a characteristic energy scale associated with the coherence sector.

The gravitational action of the theory then takes the form

$$S = \int d^4x \sqrt{-g} \left[\frac{M_{\text{Pl}}^2}{2} F(C, \partial C) R - \frac{1}{2} \sum_i \partial_\mu \phi_i \partial^\mu \phi_i - U(\phi_i, C) + \mathcal{L}_m \right], \quad (97)$$

where M_{Pl} denotes the reduced Planck mass, ϕ_i represent the five Maat fields, $U(\phi_i, C)$ is the effective potential governing their interactions, and $\mathcal{L}_m$ is the matter Lagrangian.

This formulation promotes the MFToE framework from a purely algebraic coherence modulation to a dynamical scalar–tensor response. The gradient term introduces an additional channel through which variations in the coherence field can influence the gravitational sector.

Importantly, the present work focuses on the simpler baseline limit in which the coherence field varies slowly on cosmological scales. In this regime,

$$\nabla_\mu C \rightarrow 0, \quad (98)$$

and the theory reduces to the algebraic MFToE coupling studied in the previous sections.

The dynamical extension introduced here therefore represents a natural generalization of the baseline model. In principle, it could lead to observable signatures in cosmological perturbations, structure growth, and large-scale clustering. A full analysis of these effects lies beyond the scope of the present work and will be explored in future studies.

References

- [1] Christof Krieg. Mftoe vacuum dynamics: From toy/eft integrator to desi dr2 bao comparison, 2025. Documentation and reproducibility notes.